

A “Quarto” Life Crisis

Are you still living the “Copy-Paste” lifestyle? Moving data into tables, images and screenshots into folders, and repeats each time changes are applied?

Section 1

Quarto

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- makes the process of creating and collaborating on scientific and technical documents dramatically better.
- is Posit's attempt to bring R Markdown to everyone! Unlike R Markdown, Quarto doesn't have a dependency or requirement for R.

Minimum requirements

for writing research papers in PDF



i Note

RStudio v2022.07 and later includes support for editing and preview of Quarto documents.

Section 2

Integrated workflow

Integrated workflow

R Markdown and Quarto allow for the seamless integration of narrative text, raw data, and executable code in a single document.

Simple Markdown syntax

Feature	LaTeX	Quarto
Bold	<code>\textbf{Heavy}</code>	<code>**Light**</code>
<i>Italic</i>	<code>\textit{Slanted}</code>	<code>*Slanted*</code>
Links	<code>\href{url}{text}</code>	<code>[text](url)</code>
Code	<code>\texttt{fixed}</code>	<code>`fixed`</code>
Images	<code>\includegraphics{...}</code>	<code></code>

Verdict: *Less typing, more presenting for Quarto.*

Code execution control

Option	Description
<code>eval</code>	Evaluate the code. If <code>false</code> , the code is never executed.
<code>echo</code>	Show the source code. If <code>false</code> , the code stays hidden.
<code>include</code>	The “Master Switch”. If <code>false</code> , the code runs but neither the code nor the results appear in the slide.

See other execution options.

Section 3

Dynamic updating

Dynamic updating

Results, tables, and figures update automatically when the underlying data changes, eliminating manual editing errors and ensuring the final paper remains up to date with the analysis.

Input

```
# load data
library("kableExtra")
data(iris)
iris2 <- iris[4:6, c(1, 3, 2, 4, 5)]

# calculations
mean_sepal_length <- mean(iris2$Sepal.Length)
mean_sepal_width <- mean(iris2$Sepal.Width)

# format into a pretty table
names(iris2) <- gsub('[.].+', '', names(iris2))
rownames(iris2) <- NULL
iris_table <- kable(iris2, booktabs = TRUE) %>%
  add_header_above(c("Length" = 2, "Width" = 2, " " = 1)) %>%
  add_header_above(c("Measurements" = 4, "More attributes" = 1))
```

Output

The means of sepal length and sepal width are 5 and 3.53, respectively.

Table 3: Iris Data

Measurements				More attributes
Length		Width		
Sepal	Petal	Sepal	Petal	Species
4.6	1.5	3.1	0.2	setosa
5.0	1.4	3.6	0.2	setosa
5.4	1.7	3.9	0.4	setosa

Section 4

Multi-format outputs

Multi-format outputs

One source file can be rendered into professional PDFs, Word documents or Tex files for collaborators, interactive HTML reports and presentations, as well as into beamer or PowerPoint.

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 - ii) html

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 - i) beamer
 - ii) html
 - iii) tex file

Section 5

Automated elements

Automated elements

Complex tasks such as cross-referencing (See Section 18, see Cross-referencing), bibliography management (See Section 19), journal formats (See Section 20), plots and table formatting are handled programmatically, which maintains consistency throughout long documents.

Cross-referencing

Citations

(e.g., JASA)

Extensions for journals

Section 6

Challenges

Challenges

Section 7

References

References

jpatteet. 2022. “Quarto Rendering Pdf, Running Xelatex 10 Times.” November 22, 2022. <https://github.com/quarto-dev/quarto-cli/issues/3518>.

Posit. n.d. “Comprehensive Guide to Using Quarto.” <https://quarto.org/docs/guide/>.